

# 10 Annex A Site and Environmental Conditions

ALDRIDGE HARBOR PORT AUTHORITY

## KESSLER POINT MARINE TERMINAL — PERIMETER SECURITY, ACCESS CONTROL AND SURVEILLANCE SYSTEMS

DOCUMENT 10 OF 13 — ANNEX A: SITE AND ENVIRONMENTAL CONDITIONS

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|---------------------|--------------------|
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SENSITIVE SECURITY INFORMATION — 49 CFR PART 1520

### 1. SITE

A-1.1 Kessler Point Marine Terminal is a container terminal on the United States Gulf Coast, comprising approximately **185 acres**, with a quay face of approximately **4,900 linear feet** and a landside boundary of approximately **8,200 linear feet**.

A-1.2 The terminal handles approximately **950,000 TEU** annually.

A-1.3 The terminal is a facility regulated under 33 CFR Part 105 and operates under a Facility Security Plan approved by the Captain of the Port.

A-1.4 The terminal operates continuously. Vessel operations are conducted twenty-four hours a day, seven days a week.

### 2. CLIMATIC CONDITIONS

A-2.1 The following design conditions shall be used:

| Parameter                                   | Value                                                       |
|---------------------------------------------|-------------------------------------------------------------|
| Maximum ambient dry bulb temperature        | +41 °C                                                      |
| Minimum ambient dry bulb temperature        | −5 °C                                                       |
| Maximum relative humidity                   | 100%                                                        |
| Mean annual rainfall                        | 1,580 mm                                                    |
| Maximum recorded 24-hour rainfall           | 340 mm                                                      |
| Design wind speed, 3-second gust            | 160 mph (Risk Category III)                                 |
| Seismic design category                     | B                                                           |
| Atmospheric corrosivity category (ISO 9223) | C5-M, marine                                                |
| Airborne chloride deposition                | High — site is within 500 m of open water on two boundaries |
| Solar radiation, maximum                    | 1,050 W/m²                                                  |
| Lightning flash density                     | 12 flashes/km²/year                                         |

A-2.2 The Site lies within a hurricane-exposed region. All externally mounted equipment shall remain secure at the design wind speed and shall be capable of being secured or removed in advance of a declared storm event.

A-2.3 Equipment enclosures shall be suitable for continuous exposure to atmospheric corrosivity category C5-M.

### 3. OPERATIONAL CONDITIONS

A-3.1 Gate operating hours are 06:00 to 18:00 Monday to Friday, and 08:00 to 16:00 on Saturday. Night gates operate on announcement.

A-3.2 Typical truck volumes:

| Measure                | Value                 |
|------------------------|-----------------------|
| Truck visits, weekday  | 1,500 – 2,500         |
| Peak hour truck visits | 250 – 375             |
| Annual truck visits    | approximately 500,000 |

**A-3.3** Labour shift start times are 08:00, 18:00 and 03:00. The 08:00 shift start produces a pedestrian flow of approximately **8 to 12 persons per minute** at the main personnel entry over a twenty to thirty minute window.

**A-3.4** Approximately **350 to 650** individuals enter the Site on a typical day. The access control database shall accommodate approximately **1,800** credential holders.

**A-3.5** Container stacking reaches five high in the yard and may obstruct lines of sight from the perimeter. Stack configurations change daily.

**A-3.6** Rubber-tyred gantry cranes, reach stackers, terminal tractors and empty handlers operate throughout the yard at all hours. All equipment installed at grade in the operational area shall be protected against impact.

**A-3.7** Rail operations are conducted on the eastern boundary. The rail gate is opened on demand.

## 4. GROUND CONDITIONS

**A-4.1** The Site is constructed on hydraulically placed dredged fill over soft marine clay.

**A-4.2** Ground investigation records are available for inspection at the Authority's offices. They are not exhaustive and shall not be relied upon as a complete record.

**A-4.3** Buried services within the Site are extensively documented but the records are known to be incomplete. Undocumented services shall be anticipated.

**A-4.4** Contaminated fill has been encountered in previous works in the northern sector. Disposal shall be in accordance with applicable regulation.

**A-4.5** Groundwater is encountered at approximately 1.5 m below ground level and is brackish.

## 5. EXISTING SYSTEMS

**A-5.1** The existing perimeter comprises 6 ft chain link fencing with barbed topping, in poor condition, without electronic detection.

**A-5.2** The existing video surveillance system comprises 38 analogue cameras and 6 network cameras on a proprietary recorder, with approximately 14 days of retention. The existing system shall remain in operation until the replacement system for the affected area has been accepted.

**A-5.3** The existing access control system comprises card readers at four locations on a legacy platform no longer supported by its manufacturer. Credential data shall be migrated to the new system.

**A-5.4** The existing terminal operating system is in use and shall not be modified.

**A-5.5** The existing fire alarm system is addressable and has spare capacity for the interface required at IF-6.

## 6. UTILITIES

**A-6.1** Electrical supply is available at the Security Operations Centre location at 480 V, three phase, with capacity of 400 A.

**A-6.2** Supply to field positions shall be derived from the nearest terminal distribution board. Available capacity at each is stated on the drawings.

**A-6.3** Existing duct routes are shown on the drawings. Their condition and capacity have not been verified.

**A-6.4** Potable water and foul drainage are available at the Security Operations Centre location.

## 7. DATA RETENTION AND RECORDS

**A-7.1** Records generated by the security systems shall be retained as follows:

| Record type                  | Retention period |
|------------------------------|------------------|
| <b>Recorded video</b>        | <b>45 days</b>   |
| Access control transactions  | 24 months        |
| Alarm and event log          | 24 months        |
| Operator action audit log    | 24 months        |
| Pre-gate transaction record  | 24 months        |
| Watchlist query results      | 12 months        |
| System configuration backups | Life of system   |

**A-7.2** Retention periods shall be configurable by an administrator.

**A-7.3** Records shall be exportable in a non-proprietary format for evidential purposes.

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## 8. LIGHTING

**A-8.1** Existing perimeter lighting is provided by high-mast units at approximately 400 ft centres, generally in the operational yard rather than at the fence line.

**A-8.2** Measured illuminance at the fence line ranges from 0.5 lux to 12 lux depending on proximity to a mast.

**A-8.3** Additional lighting is not within the scope of this Contract. The Contractor shall satisfy itself that its proposed system performs at the illuminance levels stated.

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## 9. ACCESS AND WORKING RESTRICTIONS

**A-9.1** Site working hours are as stated at SOW-4.5.

**A-9.2** No crane may be operated within the container yard during vessel operations without the prior agreement of the terminal operator.

**A-9.3** Work at the quay face is subject to vessel schedules and may be interrupted without notice.

**A-9.4** Hot work requires a permit issued by the Authority.

**A-9.5** In-water work is subject to environmental windows and to separate permitting. No in-water work is anticipated under this Contract.

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**END OF DOCUMENT 10**